
Coherent Excitation in Active Biological Systems

H. Fröhlich

ABSTRACT: The principal types of coherent excitations: metastable state with high electric dipole moment, and coherent high-frequency electric vibrations are derived from simple models. Far-reaching consequences for biological systems are discussed. The conjecture is supported by a great variety of experiments.

1. General

From the point of view of atomic and molecular physics (and chemistry), biological materials are extremely complex and complicated systems. Yet they function in a most systematic way. Molecular biology succeeded in establishing the structure of the relevant giant molecules. Thus, e.g., it was found that enzymes possess an active site where the relevant processes can take place according to the laws of chemistry. Yet, this fails to explain their enormous catalytic power. Clearly, function must be treated in terms of dynamic, rather than static, properties. As a next step one may be tempted to introduce the known interaction between the atoms and ions of the system and, based on the known structure, calculate their dynamical properties. Such a procedure would be relevant in the range of linear displacements from a rigid structure like a solid. Enzymes, however, do not form such a rigid structure. Attempts to carry out the appropriate calculation have been discussed recently.⁽¹⁾ Even for small enzymes they

H. Fröhlich • Department of Physics, Oliver Lodge Laboratory, Oxford Street, P.O. Box 147, Liverpool L69 3BX, England.

require quite considerable computer time. Such endeavors, thus, while useful as a demonstration of the importance of dynamic properties, will not result in a general theory of the relevant processes, if such a theory exists.

One might then try an approach that emphasizes the common features of various systems—enzymes in the above example—and treat them in a general way thus creating a framework into which the individual cases will fit. To take an example from pure physics, consider the dynamic properties of fluids. Here we do not proceed by first considering the structure of a particular fluid, a most formidable task, and then, based on this knowledge, investigate the dynamic properties. If this were possible then we would obtain the motion of individual atoms or molecules which constitute the fluid, a result which is not very interesting. What is interesting is the space-time dependence of the mean density, $\rho(x, t)$ and of the mean current density, $j(x, t)$. Both are connected by a set of differential equations, the Navier–Stokes equations, from which their space-time dependence can be derived in terms of certain parameters like compressibility, frictional constants, which define the particular fluid. These Navier–Stokes equations can be derived exactly from the basic multiparticle equation, the Schrödinger equation,⁽²⁾ not by solving it in terms of the multiparticle coordinates, but rather by first expressing ρ and j in terms of individual particle properties, and then finding the differential equations satisfied by them, which essentially arise as symmetry properties of the multiparticle system.

The success of this procedure rests in the definition of the two multiparticle quantities ρ and j , which describe the essential properties of hydrodynamic flow. Derivation of the actual values of the various parameters entering the equations of flow would require, of course, deeper investigation of the basic multiparticle equation. Could we not apply a corresponding method in the treatment of the dynamic properties of biological systems? The answer is no, not for the time being, for we do not yet know the quantities that would take a role corresponding to ρ and j in the previous case, though later in this chapter we shall make proposals in this respect and discuss their experimental verification. The difficulty arises from the enormous number of states of which excited multiparticle systems, like the biological ones, are capable and which make a systematic investigation and inspection impossible. Thus the lowest energy state of a system of N particles, the ground state, usually—though not always—is a crystal. Small displacements of the particles from their equilibrium positions can be treated in a systematic way as linear excitations. In biological systems, however, we are very far from these states, in the realm of nonlinear excitation. The number of states then rises exponentially with the number N of particles—in some particular case say as 10^N . Quite clearly it is impossible, even with a supercomputer, to evaluate and

investigate all these states. Hence we must find a way of deciding on "interesting" properties, usually connected with some kind of order. This order need not, however, be of a spatial nature. In a machine, for instance, this order consists in its way of functioning. Normally we treat this with the laws of macrophysics, which, in turn, follow from the laws of microphysics.

Early microbiologists were tempted to consider spatial order as the only order existing in physics. Thus, after no spatial order had been found in the arrangement of amino acids in an enzyme, Monod⁽³⁾ says that knowing the arrangement of 199 amino acids would not permit us to predict the 200th. He concludes that the arrangements are random. Actually, however, the whole chain must function as an enzyme; replacing the "correct" 200th amino acid by a "wrong one" disturbs this function.

In looking to physics for guidance to find a concept capable of describing biological "order" we are led to the concept of coherence, which applies to cases as different as superconductors, superfluids, lasers, and masers. Roughly speaking it implies that if a certain property is known at a space-time region near (x, t) it is also determined at another, (x', t') . Thus in a coherent wave, for instance, amplitude and phase at (x, t) determine their values at another, (x', t') . Considering now biological systems, periodic processes are very common among them with periods ranging from that of life cycles to very high-frequency electric vibrations. In contrast to most physical periodic processes, periods of biological processes often vary from case to case. Thus the periods of the life cycles of cells grown under exactly the same conditions vary within certain limits, i.e., each cell exhibits a certain amount of personality. Also, if an apparent cause for a certain biological process is removed, another takes over because biological processes are frequently multicausal.

We shall presently describe the derivation of certain coherent excitations on the basis of definite physical models. Such developments show that random energy supplied to a certain system need not lead to heating but may result in the excitation of ordered (coherent) states. This need not imply that other models could not lead to similar excitations, i.e., that a multicausal situation exists—in other words that experimental verification of the excitation need not necessarily be considered as proof of a particular model. Clearly a situation then arises which requires close collaboration between theory and experiment. Thus, e.g., theory has predicted certain coherent excitations and experiment verifies their existence. This presents a development from the point of view of physics. From the point of view of biology, however, we may ask a question that is prohibited in physics: what is the purpose of these excitations? Evidence will be presented later in this chapter for the first (physical) stage, but the second, biological question has hardly been touched yet. Its solution will, of course, lead back again to physics, i.e., a certain process has certain consequences.

If these consequences are vital biologically then multicausal possibilities must be considered again, i.e., what other processes could have the same consequences.

2. Coherent Excitations

In looking for a physical characteristic common to most biological systems, one is at once led to their dielectric properties.⁽⁴⁾ Most startling among these is the high electric field of about 10^5 V/cm present in biological membranes. This field acts in the range of nonlinear response. Materials inside membranes are, therefore, strongly, nonlinearly polarized. Furthermore, the many ions contained in biomolecules make an essential contribution to their stabilization through their interaction⁽⁵⁾ as well as through their interaction with the surrounding medium, e.g., water.⁽⁶⁾ On the other end biological systems are capable of reacting to extremely weak external impulses as reported in Section II of Ref. 7. Hence, by appropriate arrangements and excitations, biological systems may form most sensitive instruments that in normal technology would be devised in completely different ways with very different materials.

As a first case of coherent excitation, a metastable state with very high electric polarization (ferroelectric state) will be shown to exist in materials with high polarizability, combined with elastic softness, as discussed in Section III B of Ref. 7, and Ref. 8. Such excitation is coherent as the polarization at a point x determines the value at another, x' . The metastability of this excitation arises from the shape dependence of the electrostatic energy of a material. Thus, e.g., the ratio of the energies of a slab and that of a sphere both having equal volume and polarization is $3\varepsilon/(\varepsilon + 2)$, where ε is the dielectric constant (cf. p. 168 of Ref. 9). To the electrostatic energy must be added the elastic energy arising from a deformation of the material.

Thus consider a sphere with polarization P and electrostatic energy $\frac{1}{2}\gamma P^2$. If deformed into an ellipsoid with excentricity η , this energy is changed by terms proportional to η , and higher powers, as the electrostatic energy does not have an *extremum* for the sphere, $\eta = 0$. The elastic energy arising from the displacement will, however, have no linear term in η but be proportional to η^2 , as elastically the sphere is the equilibrium state. The total energy U , therefore, for small η will have the form

$$U = \frac{1}{2}\gamma P^2(1 + c_1\eta + \frac{1}{2}c_2\eta^2) + \frac{1}{4}C\eta^2 \quad (1)$$

where $C > 0$ and $\gamma > 0$, and the C_2 term has been inserted as it might influence the C term. Minimizing with regard to η yields

$$\eta = -c_1\gamma P^2/(C + c_2\gamma P^2) \quad (2)$$

and hence

$$U = \frac{1}{2}\gamma P^2 - (c_1 \frac{1}{2}\gamma P^2)^2 / (C + c_2 \gamma P^2) \quad (3)$$

Thus U rises more slowly than it would without the deformation ($\eta=0$), and it reaches a maximum beyond which it decreases with increasing P^2 , provided $c_2 > 0$, and $2c_2 < c_1^2$. For still larger P^2 , higher-order P^2 terms will be relevant, so that U goes through a minimum. Should that be at negative U , then this represents the ground state and the material is ferroelectric. Otherwise this minimum represents a metastable state with very high polarization, a quasi-ferro-electric state.

On a more formal basis we may consider a homogeneous polarization field P , energy density $\frac{1}{2}\omega^2 P^2$, interaction with an elastic field with potential energy density $\frac{1}{2}s^2 \rho^2$ for longitudinal elastic displacements yielding a density ρ . Their interaction must be proportional to ρP^2 , to provide the effect discussed above.

The total potential energy has then the form

$$W = \frac{1}{2}\omega^2 P^2 + (\frac{1}{2}s^2 \rho^2 + c P^2 \rho)(1 - d^2 P^2) \quad (4)$$

where c and d are constants and the $d^2 P^2$ term has been included to take care of nonlinearity required to stabilize the material. Minimizing with regard to ρ yields

$$\rho = -\frac{c}{s^2} P^2 \quad (5)$$

and hence

$$W = \frac{1}{2}\omega^2 P^2 - \frac{1}{2}\frac{c^2}{s^2} P^4 + \frac{1}{2}\frac{c^2 d^2}{s^2} P^6 \quad (6)$$

Thus this potential energy increases proportional to P^2 for small P^2 , and proportional to P^6 for large P^2 . In between it has a maximum, followed by a minimum provided a parameter λ satisfies

$$\lambda = 3s^2 d^2 \omega^2 / c^2 < 1 \quad (6a)$$

and this minimum represents a metastable state when $\frac{3}{4} < \lambda < 1$, while for $\lambda < \frac{3}{4}$ this highly polar state is the ground state. This confirms our previous results, and (6a), moreover, indicates the conditions the material must satisfy. They are, in particular, elastic softness (small s) and high polarizability (small ω) in conjunction with large coupling between the fields (large c^2).

Existence of highly polar metastable states should thus be a general property of biological materials. One would expect this to hold in less sim-

plified models, introducing particular structures together with anisotropy. One would expect then to find more than one metastable state. A small generalization of the above is possible, in fact, if the various parameters s , ω , c slowly depend on space coordinates, provided derivatives of P and s need not be introduced into the expression for the potential energy.

Most generally, of course, small excitations in materials with definite molecular structure are described through frequency-wave number bands, which can be classified as polar or nonpolar. Introduction of nonlinear features, as required here, must then be investigated for various particular cases, though our above treatment should always be correct as a limiting case. From the structural point of view, metastable excited states would then be classified as arising from conformational changes.

A candidate frequently mentioned in this respect is the displacement of H^+ in an H bond (e.g., Refs. 10–12 and related papers), for which also evidence from an analysis of dielectric properties in the millimeter region is available.^(13,14) Displacement of various H^+ ions does, of course, yield electric polarization, though this need not be homogeneous. It is important, however, to realize that this polarization necessarily is connected with electric polarization of the rest of these molecules, which in turn leads to elastic displacements in it, as follows from our general model. Within this frame, polarization of the H bond represents a subsystem which enhances the stability of the metastable state. Detailed calculations have not been carried out yet.

So far we have considered the potential energy W of our system only, and found it to have a minimum at $P=0$ and at $P=\pm P_0$ if P_0^2 is the value of P^2 at which the metastable state occurs. The system can of course oscillate harmonically if the displacement from $P=0$ or $P=\pm P_0$ is small. More generally, however, the system will carry out anharmonic oscillations. In fact it is shown in Ref. 8 that to a good approximation P satisfies the equation of motion

$$\ddot{P} + \frac{\partial V(P)}{\partial P} = 0 \quad (7)$$

where

$$V = A + \frac{1}{2}BP^2 + CP^4 \quad (8)$$

i.e., as long as P^4 can be neglected, the oscillations are harmonic. Yet even then the system possesses highly nonlinear properties as A , B , C are not constant but depend on the time average p of P^2 ,

$$p = \overline{P^2} \quad (9)$$

by

$$A = \frac{1}{2} \frac{c^2}{s^2} p, \quad B = \omega^2 - 2 \frac{c^2}{s^2} p - \frac{d^2 c^2}{s^2} p^2, \quad C = \frac{c^2 d^2}{s^2} p \quad (10)$$

Thus each amplitude of P defines its own potential V . Clearly for small amplitudes when p can be neglected in B , and the P^4 terms are also negligible, the frequency is ω . For growing amplitude, but when P^4 is still negligible, the frequency of oscillations is softened, i.e., B becomes smaller until finally it changes sign; V then has a maximum at $P=0$ and a minimum further outside.

Near the metastable state, P may, of course carry out small, harmonic oscillations again, but with a frequency different from ω .

The coherent excitation considered so far is essentially static; its establishment requires a supply of energy larger than the energy of the metastable state above the round state $P=0$. A dynamic, very different type of coherent excitation may exist whose maintenance requires a continuous supply of energy at a rate larger than a critical s_0 . The model is again based on polar oscillations of a large biological system, forming a band of modes with frequencies $\omega_1, \omega_2, \dots, \omega_z$. Metabolic energy is considered to be supplied at a rate s_j to mode ω_j . This systems of polar modes is considered to interact with a heat bath which is kept at a constant temperature T . In cells, this will largely consist of cell water with its ions. Linear interaction will then impose a Planck distribution for the excitation of the polar modes. This interaction is described in terms of emission and absorption of single quanta $\hbar\omega_j$ between the modes and the heat bath. When nonlinear interactions are taken into account, however, then energy exchange in terms of two quanta will occur. Most important among these are simultaneous emissions and absorption of quanta $\hbar\omega_j$ and $\hbar\omega_l$. If n_j represents the number of quanta by which mode ω_j is excited, then the following equation arises for the rate of change, $\dot{n}_j$ (cf. Section III C of Ref. 7):

$$\begin{aligned} \dot{n}_j = & s_j - \phi_j(n_j e^{\beta\omega_j} - (1 + n_j)) \\ & - \sum_l \chi_{jl}[n_j(1 + n_j) e^{\beta\omega_j} - n_l(1 + n_l) e^{\beta\omega_l}] \end{aligned} \quad (11)$$

where $\beta = \hbar/kT$, and the ϕ and χ are transition probabilities. The ratio of absorption and emission terms, proportional to n and $1 + n$, respectively, follows from the principle of detailed balance. Note that after further summation over j , the χ terms vanish so that in the stationary case when $\dot{n}_j$ vanishes, the total number $\sum n_j$ of quanta that arises from the total rate of supply $\sum s_j$ is determined by the first-order terms only through the ϕ . The

actual distribution will, however, be largely determined by the χ terms, provided that they are relatively large. In the stationary case then they must vanish so that

$$n_j = \frac{1}{\exp[\beta(\omega_j - \mu)] - 1} \quad (12)$$

follows where μ must be determined from the already known total number of quanta $\sum n_j$. When this number exceeds a critical one, i.e., when $\sum s_j$ exceeds s_0 , then μ must approach the lowest ω_1 , yielding a large n_1 , as in Bose condensation. This frequency is then very strongly, i.e., coherently, excited. Detailed considerations⁽⁷⁾ show that in certain circumstances other frequencies might be strongly excited.

To understand the possibility of this type of coherent excitation, the strong interaction with the heat bath must be emphasized. It endeavors to impose its temperature on the distribution which is demonstrated by the form (12), which represents a Bose distribution at temperature T . But while in a Bose gas the number of particles is fixed so that the Einstein condensation arises only when the temperature is sufficiently lowered, in our case the temperature is fixed, but the number of particles (quanta) increases with increasing rate of energy supply.

It must also be emphasized that this type of coherent excitation arises from the strong nonlinear coupling with the heat bath (χ terms). Other terms could be introduced, in principle, which would counteract this effect. Thus a very strong nonlinear interaction among the quanta of the polar modes, without the use of the heat bath, would simply tend to increase the temperature. We thus have found a particular example of the two different ways energy supply to a system may act, resulting in either an increase in temperature, or in the creation of a new order, that may be dynamic. Our considerations have emphasized the second possibility. Experiment must decide whether biological systems make use of it.

Finally, in this section, possible frequencies for the excitation of coherent vibrations should be considered. Originally special reference was made to membrane oscillations as their high polarization will then yield coherent electric oscillations (cf. Ref. 7, with earlier literature). Since the thickness of a membrane is about 10^{-6} cm and its elastic constant is equivalent to a velocity of sound of about 10^5 cm/s, a frequency of the order of 10^{11} Hz was expected, corresponding to electromagnetic waves in the millimeter region. When based on proteins or DNA, however, both higher and lower frequencies may be expected.

It must be emphasized at this point that most of the polar modes of a system interact only weakly with electromagnetic waves of the same frequency, as usually the wave lengths do not match. Thus, e.g., excitation of

sections of the outer membranes of a cell may result in a very high multipole electric oscillation, and hence in small interactions with electromagnetic waves of the same frequency.

3. Consequences

The two coherent excitations, metastable highly polar states, and coherent electric vibrations, may have far-reaching consequences arising from the forces exerted by the respective fields which are absent in the non-excited states. The electric field in the metastable state is static; it thus acts inside and near the excited system, while at larger distances it is screened by the ions in the surrounding medium. For molecules, e.g., proteins, surrounded by cell water, the screening distance naturally is much shorter than for proteins dissolved in membranes. In the case of excitation of coherent vibrations, on the other hand, screening is negligible once the frequency exceeds a critical one.

Forces arising from the excitation of the metastable state may then be employed to reduce activation energies in enzyme processes.^(15,4) The system enzyme + substrate will change with the motion of ions in it and, hopefully, the direction and magnitude of the internal field will change as required by the process. The whole system, enzyme + substrate, while active, possesses a high electric dipole moment which, in principle, could be measured by standard dielectric methods. Such measurements have not been attempted yet; they face difficulties arising from the presence of unattached ions in the medium in which the enzyme reaction occurs.

For a protein dissolved in a membrane the possibility arises that it may act as a switch element. For if μ is the dipole moment in the metastable state, and F is the electric field in the membrane, then if the directions of μ and F are appropriate the interaction lowers the energy by $(\mu - \mu_0)F$, where μ_0 is the dipole moment in the ground state. If this energy equals the energy of excitation of the metastable state, then the system is degenerate with two equal energy states. They then can be switched by the addition of relatively weak fields.

It should also be noticed that in proteins, aligned in a membrane, the excitation energy may be transferred from a particular excited protein to its neighbor, thus providing transport of energy that is nearly loss free.

Solitons, furthermore, represent a particular case of excitation to a metastable state.^(16,17) In the Davydov soliton, in particular, stretching of the $C=O$ bond forms the polar excitation, stabilized by the surroundings which play the part of the elastic field. This soliton, too, provides a loss-free transport of energy.

Excitation of a metastable state requires a certain minimum amount of energy. Excitation of the coherent polar mode, on the other hand, requires supply of a minimum rate of energy, s_0 ; its maintenance thus requires biological pumping. Its excitation results in a long-range, frequency-selective interaction that may lead to attraction between two, or many molecules, provided they oscillate with equal frequency. In particular cases this may also result in selective repulsion. In the case of the enzyme-substrate system, in the first stage the (frequency) correct substrate is attracted to the enzyme (long-range selectivity); in the second stage the usual short-range chemical processes take place—an example of a multicausal process.

As indicated in connection with Eq. (11), the energy supply s_j may occur at any, or all, frequencies of the polar modes. The subsequent excitation of a particular mode, e.g., ω_1 , then requires some time even when the total rate of supply exceeds s_0 . The system thus possesses storing ability. One would expect, however, that this time is particularly short if the energy is supplied at the frequency of the mode that will be excited coherently. Furthermore, arising from the considerations of mode softening, this mode, when excited, may be detached from the band and hence provide a very frequency-sensitive target for further energy supply. Detailed calculations on this question have not been carried out yet.

Consider now biological processes that make use of coherent excitation of a particular polar mode, requiring pumping at a rate $\sum s_j > s_0$. Let (cf. Section IV C of Ref. 7) t_1 be the time required for this process; then, we expect it to be the faster the larger $\sum s_j$; but it will never occur ($t_1 = \infty$) as long as $\sum s_j < s_0$. Thus if a and n are positive numbers we may assume

$$t_1 = a \left/ \left(\sum s_j - s_0 \right)^n \right., \quad \sum s_j \geq s_0 \quad (13)$$

If we now consider a complex biological process consisting of many steps, only one of which depends on the excitation, i.e., on s_j , then if t_2 is the time required for all these s_j -independent processes, the total time for the process is

$$t = t_1 + t_2 \quad (14)$$

i.e., the total rate is

$$\frac{1}{t} = \frac{(\sum s_j - s_0)^n}{a + t_2(\sum s_j - s_0)^n}, \quad \sum s_j > s_0 \quad (15)$$

It thus depends in a steplike manner on the rate of supply $\sum s_j$.

We must assume now that the biological system is controlled in such a manner that the rate of energy supply is switched on at a period that integrates the particular process considered here with the whole biological development. External supply of energy, e.g., by microwaves at a rate say s_e , will then influence the rate $1/t$ and hence interfere with the general development. The effect of even a small outside s_e can be very big at a stage when the biological $\sum s_j$ is close to s_0 , because replacing $\sum s_j$ by $\sum s_j + s_e$ in Eq. (15) is equivalent to replacing s_0 by $s_0 - s_e$, i.e., it shifts the steplike curve Eq. (15). At particular times in the biological development, small s_e hence may cause large effects, in particular when s_e is supplied at the "correct" frequency, which involves little time for the reaching of the required quasistationary distribution of the excitation of modes.

In considering now candidates that may act as carriers for the electric vibrations arising in the coherent excitation of a particular mode, recent results on the structure of cells must be presented. It has been found⁽¹⁸⁾ that an intricate network of protein structures known as the cytoskeleton exists, among which the so-called microtrabecular lattice extends from the cell surface throughout the cytoplasm. It has also been proposed⁽¹⁸⁾ that enzymes do not float in the cytoplasm but are attached to this lattice, which may act as carrier of the coherent electric vibrations that provide frequency-selective interaction between appropriately excited molecules. Supporting this conjecture is the conclusion that "microtubules ... form a vectorial framework upon which some other force generating component is organized."⁽¹⁹⁾ This also is supported by theoretical investigations on self-focusing.⁽²⁰⁾ To avoid misunderstandings it should be repeated again that while the relevant frequencies might be in the range of electromagnetic millimeter waves, interaction with such radiation will be weak as the wavelengths will not be in the millimeter region, but probably will be less than the cross section of the tubules, possibly of the order of molecular distances.

It will now be shown that combination of our two coherent excitations may result in low-frequency periodic enzyme reactions giving rise to corresponding electric vibrations—another coherent electric vibration (Ref. 21 and Section IV E of Ref. 7). Consider an enzyme process consisting of the interaction of a substrate and an excited enzyme and assume the energy thus made available to excite a nonexcited enzyme. Then if N , Z , S are the respective numbers of excited enzymes, nonexcited enzymes, and substrates, the rate of change of N is proportional to NZS with a rate constant α . Also if there exists a spontaneous decay of excitation of an enzyme, rate constant β , then

$$\frac{dN}{dt} = \alpha NZS - \beta N \quad (16)$$

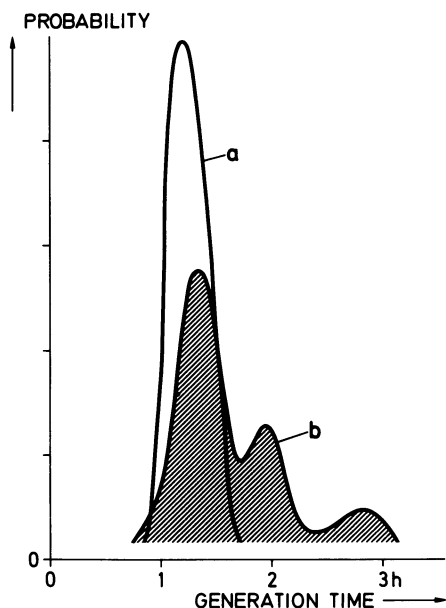


FIGURE 1. Probability of time of cell cycle when irradiated (a) with 83800 MHz or without irradiation, (b) with 83835 MHz (W. Grundler, personal communication).

and the number of substrates decreases at the rate αNZS . Consider now that during this process substrates are coherently excited to a frequency that attracts other substrates to the region (rate constant γ); then

$$\frac{dS}{dt} = -\alpha NZS + \gamma S \quad (17)$$

Consider the case that Z is so large that it can be treated as time independent; then time-independent solutions N_0 , S_0 exist,

$$N_0 = \gamma/\alpha Z, \quad S_0 = \beta/\alpha Z \quad (18)$$

Hence if

$$N = N_0 + v, \quad S = S_0 + \sigma \quad (19)$$

then

$$\dot{v} = \gamma\sigma + \alpha Z\sigma v, \quad \dot{\sigma} = -\beta v - \alpha Z\sigma v \quad (20)$$

These are the well-known Lotka-Volterra equations that have periodic solutions. In the limit that σ and v are small so that the σv term can be neglected, $(\beta\gamma)^{1/2}$ represents the frequency of oscillation, which thus is determined by the rate γ of spontaneous decay of enzyme excitation, and

by the rate β of attraction of substrate molecules. According to our first type of coherent excitation, the excited enzyme carries a large electric dipole moment, which thus also will oscillate with the frequency $(\beta\gamma)^{1/2}$. Outside electric observation will normally integrate the contribution of all enzymes and thus will also depend on their spatial disposition.

This scheme was originally derived for brain waves, and extended by taking into account a possible short-range interaction between excited enzymes, which then tends to organize them into a ferroelectric system. This results in an extension of Eqs. (19) and (20) into a complicated set of equations whose solutions are limit cycles; this means that the system will then tend to oscillate with a fixed amplitude.

This case has been investigated in detail⁽²²⁾ and it has been shown that a weak, appropriate, external perturbation may cause the collapse of the limit cycle and hence lead to the liberation of the considerable energy stored in it. This was intended as a model to demonstrate how weak electric perturbations can cause large effects in the brain (e.g., a nerve impulse).

The considerations given here are of a very general nature, and modifications of the theory can easily be performed without altering the principal result of coherent excitations. One will expect the excitation of a polar mode to be relevant whenever long-range coordination is required, as in cell division, tissue growth, and the cancer problem; cf. Section IV D of Ref. 7 and Ref. 23.

The theoretical derivation of the existence of a coherently excited polar mode following from (11) can also be phrased in a field theoretical model, or in a more exact way be supported by numerical work as quoted in Ref. 7, as well as by more recent work—the present chapter is not meant to provide a complete review. It is of importance, however, that other models involving nonlinear coupling of oscillations can lead to coherent excitation of a single mode.^(24,25)

It has also been suggested that the assembly of biological oscillators may be modeled as parameter tunnel diode memory system with couplings. The frequencies of such a system can include values very much lower than the “design” frequency, which thus supports our prediction of mode softening.⁽²⁶⁾

4. Experiments

A great variety of experiments support the conjecture of coherent excitations in biological systems. The complexity of biological materials might permit different interpretations in a number of these experiments. The possibility of understanding them from a common point of view gives strength, however, to our interpretation. In fact the conclusions of a recent

meeting⁽²⁷⁾ were "that coherent excitation had now been shown to be an intimate feature of biological activity; further progress would rely on detailed studies...." The proceedings of this meeting were published in the "Green Book."⁽²⁸⁾

A number of these experiments require subtle preparation of the biological material, which is not always described in detail and hence may lead to failure of reproducibility. As a first example consider the Raman spectrum of life algae. If certain frequencies are nonthermally excited, as required in coherent excitation, then the ratio of anti-Stokes to Stokes lines should be larger than in thermal equilibrium. This was actually found.⁽²⁹⁾ A repetition of the experiment showed no such effect.⁽³⁰⁾ The authors forgot, however, to supply CO₂, which is required to maintain the biological activity, giving rise to the excitation.

A great number of investigations on the Raman effect in active bacteria have been carried out.⁽³¹⁾ To be active, considerable dilation must be maintained, and it has been estimated that an enhancement of excitation by a factor of 10⁵ is required to obtain sufficient intensity to be noticeable.⁽³²⁾ Such enhancement cannot be expected for a single cell. A number of cells must, therefore, be assumed to oscillate coherently because the scattering intensity then is proportional to the square of the number of scattering units. While thus, seeing the lines at all already provides an indication for coherent excitation, measurement of the ratio R anti-Stokes: Stokes intensity, carried out in one case, gives a more direct proof. Thus at about 120 cm⁻⁷, in *E. coli* B bacteria, $R \simeq 1$ was measured while in thermal equilibrium $R \simeq 0.55$. Note that $R = 1$ is the largest possible value.⁽³³⁾ To maintain the required synchronization is a task that so far has not been properly documented. Adverse influences may arise from the laser beam, or from turbulence that may develop in a flow instrumentation, used to avoid destructive action of the laser beam on bacteria.

To avoid some of these difficulties a special method of synchronization has been used, whereby appropriate nutrients are provided, suddenly, after a period of starvation.⁽³⁴⁾ It was found that a nutrient line at 980 cm⁻¹ is highly excited, the enhancement above thermal being by a factor between 8 and 15; and it showed a slow periodicity of about 10 min. For a tentative interpretation we assume that nutrient molecules in or near an active cell (the measurements integrate excited and nonexcited molecules) are very highly excited and attract outside nutrient molecules to the cell, or to a particular region in it. This would then correspond to the term γS in Eq. (17). The periodicity, then, might follow in a similar way. By further investigation of the activity of this system, it should be possible to separate the diffusion part of the molecules from the region in which they are subjected to forces, and hence, possibly, determine the relevant biochemical process.

While this separation of movement due to forces, from diffusion, remains to be carried out, a very thorough separation of this type has been performed in connection with the formation of so-called rouleaux of red blood cells, i.e., stacks of cells that form in their plasma.^(35,36) It had been observed that when such cells approach to a certain distance they seem to rush towards each other. Analysis with the help of the Smoluchowski theory of diffusion revealed, quantitatively, the existence of a long-range attractive force. Its properties were found to agree with the requirements of the theory of long-range interaction arising from coherent excitation of electric vibrations in the membrane. Thus on removal of the membrane potential, the interaction disappears, but is restored when the potential is restored. Similarly it is found that the interaction disappears when the metabolism, required to maintain the excitation [terms s_j in Eq. (11)] is inhibited. Furthermore the interaction is specific in that a mixture of red blood cells from different animals segregate into rouleaux of equal origin.

It was also found that the interaction is carried by long molecules which, inside cells, corresponds to the role attributed^(18,20) to the microtrabecular lattice as discussed in Section 3. Furthermore it has been shown⁽³⁷⁾ that albumin, a globular protein, does not transmit the interaction at physiological concentrations. Albumin can, however, be converted into a transmitter by a treatment that unravels the tertiary structure and allows it to polymerize.

The frequencies for the excitation discussed here have not been measured directly, but the involvement of the membranes suggests a region near 10^{11} Hz, though both lower frequencies (mode softening) and higher frequencies (proteins dissolved in the membrane) may be contemplated. In addition, much lower frequencies arise from periodic enzyme reactions, discussed in Eqs. (16)–(20). Among other experiments that support the existence of coherent excitation⁽³⁸⁾ are the attraction of particles of a dielectric powder to active cells,⁽³⁹⁾ by the method of “microdielectrophoresis.” Investigations have been carried out on a great variety of living cells. They show that an oscillating electric field with minimum frequency of about 10 kHz exists near the surfaces of living but not of dead cells. This can be suppressed by metabolic inhibitors and is maximal (order 100 V/cm) near mitosis (in yeast).

Since the dielectric particles used in these experiments are much smaller than the cells to which they are attracted, they do not give much information about the relative phase on various sections of the cell surface. Thus a dipolar oscillation would interact strongly with electromagnetic waves of the same frequency, and the field will then decrease slowly with the distance from the cell, in contrast to the case of a high multipole whose electric field will be significant near the cell surface only. Nevertheless, within this region it will give rise to a secondary magnetic field, which

might result in the attraction of small magnetic particles, as has been observed recently.⁽⁴⁰⁾ While thus observation of emission of electromagnetic radiation would present a useful identification, lack of such observation does not imply absence of electric vibrations; for this might require measurement instruments to be placed very close to the cell surface.

An enormous literature has accumulated on the action of microwaves on biological systems, much of it, however, at single frequencies or single intensities only, which makes it difficult to analyze the results. An exception is formed by investigation of the action of coherent millimeter waves on biological systems. These wavelengths are strongly absorbed by cell water, and great care has been taken, therefore, to eliminate thermal effects by using low intensities. In this case direct action of the electromagnetic field is excluded as this would require intensities many millions of times stronger than those used in the relevant experiments.⁽⁴¹⁾ Early experiments are discussed in Section V B of Ref. 7. Their results are best summarized as follows⁽⁴²⁾: “(a) the effect of irradiation depends strongly on the frequency of the microwaves; (b) in certain microwave-power ranges, the effect of exposure depends weakly on variation of the power through several orders of magnitude; (c) the effects are observed to depend significantly on the time of irradiation.”

These results seem to confirm all theoretical predictions first made in 1968⁽⁴³⁾: As shown in previous sections, the excitation of coherent membrane oscillations should be in the 10^{11} Hz region at very sharp frequencies; the response should be steplike as a function of the supplied energy [cf. Eq. (15)], and the system has storing capability. However, the supplied energy, in the theory, is largely of metabolic nature—this has to be so as the energy supplied externally alone would be much too small to cause big effects. It has been pointed out, however, in the discussion following Eq. (15) that at an appropriate instant of the development of the biological system, a small external energy supply s_e may shift the position of the step in the response function, i.e., it will act strongly at a very sensitive stage of development. This may be compared to a phase transition in the development of the whole biological system; the externally supplied energy then changes its timing of this process relative to other processes, and the consequences for the whole system may be specific in particular cases.

It seems obvious that such subtle interference might be influenced by the natural fluctuations present in highly excited systems, like biological ones. This is in particular so as it has been shown that noise may cause phase transitions in nonlinear systems.⁽⁴⁴⁾ It might be suggested, therefore, that details of the experimentation are relevant for the outcome of the investigation.

Consider now, in particular, the sharp resonances in the rate of growth of yeast under the influence of microwaves in the 42-GHz

region.^(45,46) A great number of very sharp maxima and minima in the rate of growth versus frequency curve was found. The experiment emphasizes high-frequency stability and provides the use of two very different electrodes to supply the radiation to the yeast cultures that grow in liquid suspension. Cross correlation between the two geometries in the range between 41700 and 41800 MHz reveals a central line of about 10 MHz width with three side bands on each side, separated by about 18 MHz. The effects are relatively small, about $\pm 10\%$, but very well resolved and reproducible. Another feature, apart from the extremely sharp resonances, lies in the absence of a difference in magnitude between the two electrodes, although their surfaces differ by a factor of 10, and the radiation penetrates from the surface only a fraction of a millimeter. The suspension is constantly stirred, and it can be concluded that a single yeast cell is at most affected once during its life cycle, and then only in a particular state of development.

Clearly one would expect such extraordinary effects to arise from important, highly sensitive properties which will not reveal themselves in population effects, though these have the property of high reproducibility.

An investigation of the time of a life cycle of individual cells, and the influence of millimeter waves on it, was therefore undertaken. Frequencies in the 83-GHz region were used—for organizational reasons different from those of the source used in the experiments described above. The cell cycle time was not influenced by radiation of 83800 MHz, but very strongly so by the neighboring 83835 MHz, as shown in Figure 1. Remarkably, however, about half of the cells are not affected, whereas the others are affected very strongly. This leaves the question open whether cells show a "personality" by responding to one or another frequency, or whether a secondary effect, like a particular fluctuation at a critical time, is required to produce the effect, as suggested above.

Further experimentation should result in a more detailed knowledge of the particular cell processes involved. The sharpness of the resonances makes it quite clear, already at the present stage, that coherent excitations must be involved, and the fine structure of 18 MHz might arise from periodic enzyme reactions, Eq. (20).

While the above results are highly reproducible, it is not yet quite clear which particular items in the instrumentation, and the handling of the biological materials, are vital. Thus, at the same laboratory, a similar instrumentation has been produced which shows no effect. Up to the time of writing, the reason for this had not yet been discovered.

Other investigations in the field are published in Ref. 28. It should be expected that multicellular systems might show a larger effect. In fact⁽⁴⁷⁾ the growth of cress roots can be completely stopped within 100 seconds after irradiation with low-intensity 56-GHz microwaves, provided the

polarization is parallel to the root. From the theoretical discussion it can be concluded that the process, which owing to the effect of external energy supply is started at an inappropriate time, completely disrupts the growth. Further biological and physical investigations are required to find this process. Clearly it must, at its own time, make use of coherent excitation.

Other biological systems exist, however, whose dynamic properties are much better known than those of cress root. Among these, properties of the membranes systems of electron transport phosphorylation have been studied from the point of view of bioenergies.⁽⁴⁸⁾ This requires energy transport over relatively long distances, and other concepts that can all be supplied by various coherent excitations. Measurements to confirm this are now planned.

Finally, to return to the metastable highly polar state of individual molecules, it should be possible, in sufficiently strong external fields, to lift the molecules into this state. Transitions into a metastable state were actually discovered when strong fields were applied to so-called Langmuir-Blodgett multilayers and the low-frequency dielectric response was then measured.⁽⁴⁹⁾ Haemoglobin and some other molecules were investigated. When the applied field reached about 10^5 V/cm, characteristic changes in the dielectric response were observed. They are maintained even when they are remeasured in low fields, i.e., the layers have been transferred into a metastable state. This state persists for up to six weeks. The gradual reversion to the initial characteristics can be accelerated by heating for short periods to about 40°C . It is to be expected that, owing to the presence of water and ions, the high polarization is screened. The fact, however, that the metastable state is reached through the application of very high fields indicates its basic high polarity.

Another support for the existence of a highly polar metastable state arises in the so-called electret method. Here the material is polarized in strong fields, and the polarization is frozen in by cooling to very low temperatures. Gradual increase in temperature then releases polarization at characteristic temperatures. In our metastable state, the release should begin at temperatures above physiological ones. Both investigated enzymes, trypsin and urease, show this effect,⁽⁵⁰⁾ though the polarizing field amounted to only 25000 V/cm.

In a very different kind of experiment it was found that laser irradiation increases the activity of chymotrypsin by a factor of up to 3.⁽⁵¹⁾ This was investigated as arising from lifting the enzyme into the active, metastable state through the Raman effect. Detailed investigations of the dependence on the density of molecules and on laser power support this interpretation. For other possibilities see Ref. 52.

In summary it is seen that a variety of different experiments support the conjecture of two types of coherent excitation: a metastable state with

large electric polarization, and existence of coherent electric vibrations. It should be emphasized again that the theoretical models show that such excitations should be possible. They are not unique; other models might achieve similar results. In fact if, as we think, biological activity makes great use of these excitations, the manner of establishing them need not be unique.

References

1. Roger H. Pain, "Dynamic Proteins," *Nature* **305**, 581–582 (1983).
2. H. Fröhlich, "The Connection Between Macro- and Microphysics," *Riv. Nuovo Cimento* **3**, 490–534 (1973).
3. J. Monod, *Chance and Necessity*, transl., Collins, London (1972), p. 95.
4. H. Fröhlich, "The Extraordinary Dielectric Properties of Biological Materials," *Proc. Natl. Acad. Sci. USA* **72**, 4211–4215 (1975).
5. A. Wada and H. Nakamura, "Nature of the Charges Distribution in Proteins," *Nature* **293**, 757–758 (1981).
6. A. Warshel, "Electrostatic Basis of Structure–Function Correlation in Proteins," *Acc. Chem. Res.* **14**, 284–290 (1981).
7. H. Fröhlich, in *Advances in Electronics and Electron Physics* (L. Marton, ed.), Academic, New York (1980), Vol. 53, pp. 85–152.
8. H. Fröhlich, "Collective Behaviour of Non-Linearly Coupled Oscillating Fields," *J. Collect. Phenom.* **1**, 101–109 (1973).
9. H. Fröhlich, *Theory of Dielectrics*, second edition, Oxford Press, Oxford (1958).
10. G. Zundel, in *The Hydrogen Bond—Recent Developments in Theory and Experiment* (P. Schuster, G. Zundel, and C. Sandorfy, eds.), North-Holland, Amsterdam (1976), Vol. 2, pp. 683–766.
11. W. Kristof and G. Zundel, "Structurally Symmetrical, Easily Polarizable Hydrogen Bonds Between Side Chains in Proteins III," *Biopolymers* **19**, 1753–1769 (1980).
12. P. Rastogi, W. Kristof, and G. Zundel, "Easily Polarizable Proton Transfer Hydrogen Bonds Between the Side Chains of Histidine and the Carboxylic Acid Groups of Glutamic and Aspartic Acid Residues in Proteins," *Int. J. Biol. Macromol.* **3**, 154–158 (1981).
13. L. Genzel, F. Kremer, A. Poglitsch, and G. Bechtold, "Relaxation Processes on a Picosecond Time Scale in Hemoglobin and Poly (L-Alanine) Observed by Millimeter-Wave Spectroscopy," *Biopolymers* **22**, 1715–1729 (1983).
14. L. Genzel, F. Kremer, A. Poglitsch, and G. Bechtold, in *Coherent Excitations in Biological Systems* (H. Fröhlich and F. Kremer, eds.), Springer-Verlag, Berlin (1983), pp. 58–70.
15. H. Fröhlich, "Long Range Coherence and the Action of Enzymes," *Nature* **228**, 1093 (1970).
16. A. S. Davydov, "Solitons, Bioenergetics, and the Mechanism of Muscle Concentration," *Int. J. Quantum Chem.* **16**, 5–17 (1979).
17. H. Bilz, H. Büttner, and H. Fröhlich, "Electret Model for Collective Behaviour of Biological Systems," *Z. Naturforsch.* **36b**, 208–212 (1981).
18. J. Clegg, in *Coherent Excitations in Biological Systems* (H. Fröhlich and F. Kremer, eds.), Springer-Verlag, Berlin (1983), pp. 162–175.
19. J. Hymans, "Hooked on Microtubules," *Nature* **291**, 107–108 (1981).
20. E. Del Giudice, S. Doglia, and M. Milani, in *Coherent Excitations in Biological Systems* (H. Fröhlich and F. Kremer, eds.), Springer-Verlag, Berlin (1983), pp. 123–129.

21. H. Fröhlich, "Possibilities of Long- and Short-Range Electric Interactions of Biological Systems," *Neurosci. Res. Program Bull.* **15**, 67–70 (1977).
22. F. Kaiser, in *Coherent Excitations in Biological Systems* (H. Fröhlich and F. Kremer, eds.), Springer-Verlag, Berlin (1983), pp. 128–133.
23. H. Fröhlich, "Coherent Electric Vibrations in Biological Systems and the Cancer Problem," *IEEE Trans. Microwave Theory Tech.* **26**, 613–617 (1978).
24. H. Fröhlich, "Conditions for Coherent Excitations in Biological Systems," *Phys. Lett.* **93A**, 105–106 (1982).
25. E. Fermi, J. R. Pasta, S. M. Pasta, and S. M. Ulam, in *Collected Works of Fermi*, Vol. 2, University of Chicago Press, Chicago (1965), pp. 978–988.
26. B. S. Thornton, "Solid State Memory Problems Support Mode-Softening for Larger Assemblies of Biological Dipole Oscillations," *Phys. Lett.* **102A**, 77–79 (1984).
27. F. Keilman and D. Kell, "Coherent Excitation in Biology," *Nature* **301**, 656–657 (1983).
28. *Coherent Excitations in Biological Systems* (H. Fröhlich and F. Kremer, eds.), Springer-Verlag, Berlin (1983).
29. F. Drissler and R. MacFarlane, "Enhanced Anti-Stokes Raman Scattering from Living Cells of Chlorella Pyrenoids," *Phys. Lett.* **69A**, 65–67 (1978).
30. S. Kimoshita, K. Hirato, and T. Kushida, "Population of Vibrational State of Carotenoid Molecules in Living Cells of Chlorella," *J. Phys. Soc. Jpn* **49**, 314–321 (1980).
31. S. Webb, "Laser Raman Spectroscopy of Living Cells," *Phys. Rep.* **60**, 201–224 (1980).
32. H. Fröhlich, in *Physics as Natural Philosophy* (A. Shimony and H. Feshbach, eds.), MIT Press, Cambridge, Massachusetts (1982), pp. 287–293.
33. S. Webb, M. Stoneham, and H. Fröhlich, "Evidence for Non-Thermal Excitation of Energy Levels in Active Biological Systems," *Phys. Lett.* **63A**, 407–408 (1977).
34. F. Drissler and L. Santo, in *Coherent Excitations in Biological Systems* (H. Fröhlich and F. Kremer, eds.), Springer-Verlag, Berlin (1983), pp. 6–9.
35. S. Rowlands, in *Coherent Excitations in Biological Systems* (H. Fröhlich and F. Kremer, eds.), Springer-Verlag, Berlin (1983), pp. 154–161.
36. S. Rowlands, C. Eisenberg, and L. Sewchand, "Contractiles: Quantum Mechanical Fibrils," *J. Biol. Phys.* **11**, 1–4 (1983).
37. S. Rowlands, L. Sewchand, and L. Skibo, "Conversion of Albumin into a Transmitter of the Ultra Long-Range Interaction of Human Erythrocytes," *Cell Biophys.* **5**, 197–203 (1983).
38. H. Fröhlich, "Evidence for Coherent Excitation in Biological Systems," *Int. J. Quantum Chem.* **23**, 1589–1595 (1983).
39. H. Pohl, in *Coherent Excitations in Biological Systems* (H. Fröhlich and F. Kremer, eds.), Springer-Verlag, Berlin (1983), pp. 199–210.
40. M. Akhalaya, M. Kakiasvili, and K. Zakaraya, "On Biomagnetism of Cells," *Phys. Lett.* **101A**, 367–370 (1984).
41. H. Fröhlich, "What are Non-Thermal Electric Biological Effects?," *Bioelectromagnetics* **3**, 45–46 (1982).
42. N. Devyatkov, "Influence of Millimeter-Band Electromagnetic Radiation on Biological Objects," *Sov. Phys. Usp.* **10**, 568–569 (1974).
43. H. Fröhlich, "Long Range Coherence and Energy Storage in Biological Systems," *Int. J. Quantum Chem.* **2**, 641–649 (1968).
44. L. Arnold, W. Horsthemke, and R. Lefever, "White and Coloured Noise and Transition Phenomena in Nonlinear Systems," *Z. Physik. B* **29**, 367–373 (1978).
45. W. Grundler, F. Keilmann, V. Putterlik, L. Santo, D. Strube, and I. Zimmerman, in *Coherent Excitations in Biological Systems* (H. Fröhlich and F. Kremer, eds.), Springer-Verlag, Berlin (1983), pp. 21–37.

46. W. Grundler and F. Keilmann, "Sharp Resonances in Yeast Poove Nonthermal Sensitivity to Microwaves," *Phys. Rev. Lett.* **51**, 1214–1216 (1983).
47. F. Kremer, A. Poglitch, L. Santo, D. Sperher, and L. Genzel, "Low Intensity Millimeter Waves Inhibit the Growth of Cress Roots," *Z. S. Naturf. C.* (in print).
48. D. Kell and G. Hitchens, in *Coherent Excitations in Biological Systems* (H. Fröhlich and F. Kremer, eds.), Springer-Verlag, Berlin (1983), pp. 178–198.
49. J. Hasted, S. Husain, A. Ko, D. Rosen, E. Nicol, and J. Birch, in *Coherent Excitations in Biological Systems* (H. Fröhlich and F. Kremer, eds.), Springer-Verlag, Berlin (1983), pp. 71–83.
50. S. Mascarenhas, "Electrets in Biophysics," *J. Electrostat.* **1**, 141–146 (1975).
51. N. Kolas and W. Melander, "Laser-Induced Stimulation of Chymotrypsin Activity," *Phys. Lett.* **57A**, 102–104 (1976).
52. J. Pokorny, *J. Theor. Biol.* **98**, 20–27 (1982).